

MOHAMMED NASR EL-DIN

Electronics and Communication Engineering Student

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SUMMARY

Analog IC design engineer with a strong foundation in analog IC Design, semiconductor devices, and CMOS circuit design. Skilled in circuit analysis and simulation using Cadence/Virtuoso and Xschem/ngspice. Also experienced in machine learning and data analysis using Python.

EDUCATION

Bachelor of Electronics and Communication Engineering
Ain Shams University

Expected Graduation year 2028

TECHNICAL SKILLS

- **Analog & Mixed-Signal Design:** Analog IC Design, CMOS Fundamentals, Circuit Analysis, Logic Design
- **EDA & Simulation Tools:** Cadence/Virtuoso, xschem/ngspice, MATLAB/Simulink, Proteus
- **PCB Design:** Schematic Capture & PCB Layout
- **Programming & Data Tools:** C++, Python (Pandas, Plotly, ML Model Development), VS Code, Google Colab

PROJECTS

- **Two-Stage Rail-to-Rail CMOS Op-Amp**
Full-custom design and simulation of a two-stage 65nm CMOS operational amplifier with a floating class-AB output stage, verified across process corners, with full DC, AC/stability, noise, and transient verification in Cadence Virtuoso/Spectre.
- **Fully Differential Folded Cascode OTA**
Designed and simulated a folded cascode OTA (GF180nm), with full CMFB, AC/STB/Transient verification in Cadence.
- **Two-Stage Miller OTA Design**
Designed and simulated a two-stage Miller OTA (180nm CMOS, g_m/I_D methodology) with full open-loop, closed-loop, stability, and noise analysis verifying gain, CMRR, phase margin, slew rate, and settling time.
- **Audio Amplifier**
Designed and implemented a PCB-based analog signal conditioning circuit to amplify low-level input signals while reducing noise through filtering techniques.
- **Solar Tracker**
Designed, simulated, and implemented an analog control system for a variable-speed DC motor to control solar tracker motion, and developed a 3D-printed prototype. **Role:** Circuit Design and Simulation.
- **Electronic Lock**
Designed and implemented a digital electronic lock using logic circuits and PCB design techniques.

TRAININGS & CERTIFICATIONS

Analog IC Design ITI  summer training

summer 2026

Studied analog IC design concepts such as feedback, stability, compensation, noise analysis, and mismatch. Applied systematic analog CMOS circuit design methodologies such as g_m/I_D methodology, and practical experience with industry-standard EDA/CAD tools for circuit analysis and design.

Machine Learning Basics Workshop 

Mar 2026

Performed data preprocessing, analysis, and visualization using Python and Pandas. Developed and evaluated machine learning models for classification and regression tasks. Applied data-driven techniques to extract insights and support problem-solving.

Project: Bank Credit Score Prediction Model – Developed a ML model. Performed data preprocessing, feature engineering, model training, and evaluation.